



Zongmin Zhang

HKUST Computer Science Undergraduate | AI for Science Researcher

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EDUCATION

The Hong Kong University of Science and Technology (HKUST) 09/2023 – Present

BEng in Computer Science (Minor in Chemistry), Department of Computer Science and Engineering (CSE), School of Engineering (SENG)

CGA: 3.999/4.300 (top 2%) | Major CGA (MCGA): 4.094/4.300 | Dean's List ×5 | HKSAR Government Scholarship ×3

Selected A+ Coursework:

Chemistry: General Chemistry I & II, Organic Chemistry, Undergraduate Research.

CS & AI: OOP (C++/Java), Data Structures, Algorithms, Game Programming.

Math & Physics: Multivariable Calculus, Matrix Algebra, Statistics, Probability, Quantum Information.

PUBLICATIONS

- [1] **Zongmin Zhang**, Yuyang Lou, Bowen Zhang, Junwu Chen, Ryo Kuroki, Xuan Vu Nguyen, Edwin Fako, Lixue Cheng, and Philippe Schwaller. “AdsMind: A Physics-Grounded Multi-Agent System for Self-Correcting Discovery of Adsorption Configurations on Heterogeneous Catalyst Surfaces.” *arXiv preprint arXiv:2606.19152*, 2026. Under peer review at **Machine Learning: Science and Technology**. **First author.** Led the implementation, benchmark design and cross-backend evaluation of AdsMind, a closed-loop, physics-grounded multi-agent system for autonomous adsorption-configuration discovery on heterogeneous catalyst surfaces.
- [2] Zhilong Song, **Zongmin Zhang**, and Lixue Cheng. “Autonomous Heterogeneous Catalyst Discovery with a Self-Evolving Multi-Agent Digital Twin.” *arXiv preprint arXiv:2606.05050*, 2026. Under peer review at **Nature Communications**. **Second author.** Conducted Codex-based ablation experiments and contributed to the evaluation of CatDT, a self-evolving multi-agent digital-twin framework for autonomous heterogeneous catalyst discovery.
- [3] Aritra Roy et al. (including **Zongmin Zhang**). “From Knowledge to Action: Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry.” *arXiv preprint arXiv:2605.03205*, 2026.
Large-author community report on LLM-enabled scientific workflows from an international hackathon at the intersection of large language models, materials science, and chemistry.

RESEARCH EXPERIENCE

Undergraduate Research Assistant (UGRA), AI for Physical Sciences (AI4PhysSci) 02/2026 – Present

Lab, Department of Chemistry, School of Science, HKUST

Supervisor: Prof. Sherry Lixue Cheng

- Conducting AI for Science research on chemistry-oriented machine learning, multi-agent systems (MAS)

and safety-aware autonomous scientific discovery, leading to preprints on AdsMind, CatDT, and ChemTriage (in prep.).

Developments and Applications of Orbital-Based Learning as a General and Accurate Property Predictor 02/2026 – 05/2026

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof.

Sherry Lixue Cheng

- Investigated orbital-based machine learning representations and benchmarked baseline architectures to build general and accurate property predictors for chemical systems.

Seeing Surfaces Differently: Agentic Simulation of Atomic Rearrangement and Surface Reactivity 09/2025 – 01/2026

Semester Project, Laboratory of Artificial Chemical Intelligence (LIAC), ISIC, EPFL

/ Supervisors: Prof. Philippe Schwaller & Dr. Edwin Fako

- Developed a modular framework integrating LLM reasoning backends (e.g., Gemini, OpenRouter, Ollama) with atomistic simulation engines (e.g., MACE, OpenMD).
- Built automated conformer placement and relaxation toolchains, and curated a 20-surface benchmark dataset spanning bimetallic alloys and metal oxides (e.g., Mo₃Pd, CuZnO), establishing the prototype for AdsMind.

Assess User Experience to Design Effective Visual Representation and Interaction in Virtual Reality 02/2025 – 05/2025

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof.

Xiaojuan Ma

- Developed a multi-user collaborative VR ideation environment in Unity 6 (URP) for Meta Quest 3, resolving legacy-to-URP shader incompatibilities and texture bindings.
- Implemented real-time bimanual grab, release, and wall auto-attachment using Meta XR Interaction SDK and custom C# scripts, synchronized across headsets via Unity Netcode for GameObjects (NGO).

Knowledge Discovery over Databases 06/2024 – 12/2024

HKUST Undergraduate Research Opportunities Program (UROP) | Supervisor: Prof.

Raymond Chi-Wing Wong

- Implemented a tensor-level PyTorch data cache using vector search utilities to eliminate redundant latent vector re-evaluations for recurring item IDs, significantly reducing deep neural network (DNN) model training overhead.
- Trained an MLP autoencoder to compress high-dimensional embedding vectors from 128D to 8D with a cosine similarity of ~ 0.9 and integrated warm-up and decay learning rate schedulers to stabilize optimization.

HONORS & AWARDS

HKSAR Government Scholarship Fund – Scholarship for Outstanding Performance AY 2024/25 – 2026/27 (HK\$80,000 p.a.)

HKUST Study Abroad Funding Support

- HKUST Study Abroad Grant (HK\$10,000) Fall 2025
- HKSAR Government Scholarship Fund – Reaching Out Award (ROA) AY 2025/26

Hong Kong, China–Asia-Pacific Economic Cooperation Scholarship AY 2024/25 – 2026/27

HKUST SENG Dean's List (all semesters at HKUST) Fall 2023 – Spring 2026

EXCHANGE & VISITING

HKUST SENG Undergraduate Outbound Exchange Student	09/2025 – 02/2026
<i>École Polytechnique Fédérale de Lausanne (EPFL), School of Computer and Communication Sciences (IC), Lausanne, Switzerland</i>	
Association of Pacific Rim Universities (APRU) Virtual Student Exchange (VSE) Program	02/2025 – 06/2025
<i>Shanghai Jiao Tong University (SJTU), School of Materials Science and Engineering (MSE), Shanghai, China</i>	
Relevant coursework: MSE2602 Materials Chemistry.	

INDUSTRY EXPERIENCE

Computer Vision Algorithm Intern	06/2025 – 08/2025
<i>Sansen Well (Shanghai) Robotics Co., Ltd., Shanghai, China</i>	
- Independently built an end-to-end multi-view 3D reconstruction pipeline in C++ and OpenCV for industrial wire-rope inspection. - Implemented arcsin cylindrical unwrapping, dynamic programming seam-finding with Laplacian blending for seamless 360° texture stitching, and VTK-based 3D mesh rendering.	

TEACHING & SERVICE

IOP Trusted Reviewer, <i>Machine Learning: Science and Technology</i> (IOP Publishing);	2026
Reviewer, ICML 2026 AI for Science Workshop & NeurIPS 2026 AI for Science Workshop (OpenReview)	
HKUST CSE Programming Commons Undergraduate Teaching Assistant (UGTA)	Fall 2024, 2026
HKUST COMP2011 Undergraduate Teaching Assistant (UGTA)	Spring 2025
Student Representative, HKUST SENG Mainland Undergraduate Recruitment Trip –	06/2026 – 07/2026
JEE Main Round Interview Mingling Sessions, Guangzhou, China	

COMMUNITY & LEADERSHIP

Co-Founder & Vice President (VP), HKUST AI for Chemistry Club	09/2026 – Present
(Department-Associate Group (DAG) under the Department of Chemistry)	
Initiated a student-led academic community for seminars, reading groups, workshops, and peer learning at the intersection of artificial intelligence and chemistry.	
Member & Participant, Hong Kong Tsingtao Association (HKQDA) & Mainland	09/2023 – Present
Students and Scholars Society, UG Section (MSSSUG), HKUST	
Member & Participant, Chinese Students and Scholars Association in Lausanne (CSSA	09/2025 – 02/2026
Lausanne) & Crazy Hikers	
First Prize, Outstanding Volunteer & Professional Skill Certificate, 5th National	05/2026
University Students' Data Analysis Popular Science Knowledge Competition (Theory Track)	
Finalist Award (Team Category), HKUST GenAI Hackathon Competition on a	03/2026
Putonghua Web Tool	

ADDITIONAL INFORMATION

Technical Stack:

Languages & Systems: Python (proficient), C/C++ (proficient), Java (proficient), Rust, C#, Scala, R, JavaScript, HTML, LaTeX, Markdown, Linux/Shell scripting, OS internals, kernel-level programming

Deep Learning & ML: PyTorch, TensorFlow, CUDA, TensorRT, NumPy, Keras, scikit-learn

AI Agents & LLMs: LangGraph, LangChain, prompt engineering, RAG, Model Context Protocol (MCP), agent tool & skill design (e.g., Codex and Claude Code), harness engineering

AI4Science & Cheminformatics: RDKit, Atomic Simulation Environment (ASE), MACE-MP

Vision, Graphics & XR: OpenCV, YOLO, SAM, 3D Reconstruction, Visualization Toolkit (VTK), Unity, Meta XR SDK

Language Proficiency: Mandarin (Native), English (Full professional proficiency; Gaokao English: 142/150), Japanese (Limited working proficiency; approx. JLPT N3 level), French (Elementary proficiency; CEFR A1 level), Cantonese (Elementary proficiency)

Professional Affiliations: Student Member of ACS, RSC, CCS (Chemistry); IEEE, ACM, 2026 – Present
CCF, HKCS (Computer Science)

Research Ethics: Responsible Conduct of Research (UROP), CITI Program, HKUST Valid until 06/2029